

AMITH KUMAR SINGH

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SUMMARY

Applied AI & Research Engineer with a Ph.D. in Computer Science and 8+ years of experience spanning software engineering, machine learning research, and production-oriented AI systems. Experienced in designing, implementing, and evaluating ML and Generative AI systems, with expertise in probabilistic modeling, graph learning, retrieval-augmented generation, and scalable AI pipelines from research through deployment.

SKILLS

- Programming: Python, Java, SQL (Certified)
- Machine Learning & AI: PyTorch, Scikit-learn, RAG, GraphRAG, LLMs, Prompt Engineering, LangChain, LlamaIndex, LangSmith, RAGAS, Tool Calling, MCP
- Graph Learning & NLP: Graph Neural Networks (GCN/GAT), Knowledge Graphs, Network Science, Semantic Search, Entity Extraction (spaCy/NER), Embeddings, Hybrid Retrieval, Cross-Encoders, Rank Fusion
- Research & Evaluation: Experimental Design, Model Evaluation, Statistical Learning, Probabilistic Modeling, Hypothesis Testing, A/B Testing, Retrieval Evaluation, Error Analysis, Benchmarking
- Software Engineering: FastAPI, Flask, REST APIs, Docker, PySpark, ETL Pipelines, Structured Logging, Reproducible Workflows
- Cloud & Infrastructure: Azure AI Services, AWS (basic)

PROJECTS

GraphRAG Financial Intelligence Platform

- Built an end-to-end GraphRAG platform over SEC 10-K filings integrating vector search and knowledge graph reasoning to support multi-hop financial question answering.
- Designed a hybrid retrieval architecture combining semantic search, graph traversal, reranking, and rank-fusion techniques for entity-centric financial analysis.
- Implemented a scalable entity and relation extraction pipeline using classical NER and LLM-based enrichment for knowledge graph construction.
- Developed production-oriented APIs and retrieval services with semantic caching, structured logging, latency monitoring, and retrieval orchestration.
- Developed reusable evaluation and benchmarking pipelines using RAGAS, Recall@K, answer relevance, and latency metrics to compare retrieval strategies, identify failure modes, and support iterative model improvement.
- Benchmarked GraphRAG against vector-only RAG, achieving +12% improvement in Recall@10, 15–19% reduction in retrieved context tokens, with median retrieval latency of ~280ms per query.

Statistical Learning for Network Diffusion Models (PhD Dissertation)

Tech: Python, PyTorch, NumPy, NetworkX, Statistical Learning, Maximum Likelihood Estimation, Graph Theory

- Developed a probabilistic framework for modeling opinion diffusion over attributed networks, where node state transitions depend on neighborhood structure, node attributes, and graph topology.
- Formulated an opinion update rule with Markovian dynamics and established theoretical identifiability conditions for consistent parameter estimation.
- Derived sample complexity and statistical accuracy guarantees for parameter recovery, providing theoretical support for inference beyond predictive performance.
- Designed an approximate heuristic estimator to efficiently recover model parameters in computationally challenging settings.
- Evaluated diffusion behavior across multiple graph families including random graphs, stochastic block models (SBM), and scale-free networks to analyze how topology influences information propagation.

Probabilistic Cascade Propagation Model

- Developed a probabilistic extension of the Independent Cascade Model (ICM) for learning cascade propagation from observed diffusion data.

- Implemented SGD-based Maximum Likelihood Estimation to infer propagation parameters from over 3,500 real-world political cascade samples.
- Achieved approximately 30% improvement in precision and recall compared with the classical ICM.
- Validated the estimator through statistical confidence analysis, obtaining 95% confidence in learned parameters.

Explainable Credit Scoring Engine

- Built an interpretable, API-driven credit scoring system emphasizing fairness and transparency.
- Achieved 10% F1-score improvement over baseline models.
- Integrated explainability components and regulatory-aware thresholds (provisional patent under review).
- Designed evaluation protocols for bias, stability, and calibration.

PROFESSIONAL EXPERIENCE

ML Research Engineer - The research Foundation for SUNY (May 2022 - May 2026)

(Research Project Assistant)

- Designed and implemented machine learning, NLP, and Generative AI systems for sentiment analysis and opinion dynamics using Python, and SQL.
- Developed reproducible machine learning pipelines covering data ingestion, feature engineering, model training, evaluation, experimentation, and deployment.
- Designed Graph Neural Network architectures (GCN and GAT variants) for modeling information propagation and opinion dynamics on graph datasets containing approximately half a million nodes.
- Applied probabilistic modeling, statistical estimation, and hypothesis testing to study information diffusion and network behavior.
- Conducted experimental evaluation, benchmarking, uncertainty estimation, and model failure analysis to validate model performance and guide iterative improvements.
- Developed REST-based research services using Flask and Streamlit to expose ML and LLM models for interactive experimentation.
- Implemented reproducible experimental pipelines to benchmark competing modeling approaches through controlled experimentation and statistical analysis.
- Containerized ML systems using Docker to enable reproducible deployment and experimentation.

University at Albany (SUNY) - Teaching Assistant (Applied Computer Science) (January 2019 – May 2022)

- Supported coursework in Python, Java, and Data Structures focusing software design and algorithmic thinking.
- Reviewed code, debugged implementations, and advised on maintainable and scalable design patterns.

Nimesha Ingenieure Pvt Ltd - Senior Software Engineer (April 2017 - July 2017)

- Performed data analysis and exploration using python to identify best model features.
- Analyzed structured datasets (SQL) to generate internal reports used by engineering teams for decision-making.

IBM (SunTrust Bank) - Software Developer (April 2016 - March 2017)

- Queried and validated large-scale transactional datasets using DB2/SQL in financial systems.
- Investigated production issues, performed data-level debugging, and supported system reliability.

Cognizant (Discover Financial Services) - Programmer Analyst (September 2011 - September 2015)

- Led data migration, validation, and reconciliation for POS and payment systems using DB2/SQL.
- Supported system integration and troubleshooting across multiple environments.

EDUCATION

Ph.D. in Computer Science (GPA: 3.8) - University at Albany (SUNY), 2019 - Apr 2026

- Focus: Machine Learning, Deep Learning and Network Theory

MS in Computer Science- University at Albany (SUNY), 2017 - 2019

CERTIFICATIONS: Azure AI Fundamentals, Azure AI Engineer Associate (AI-102) (in-progress), SQL Certification (2025)